Universitas Pandrosion: Paper 105
Lehmer's Conjecture: a Pandrosion-Energy Attack

Computer-assisted verification at $d \leq 30$ on integer polynomials

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Abstract

Lehmer's conjecture (1933) asks whether the Mahler measure $M(P)$ of a non-cyclotomic monic integer polynomial admits a uniform lower bound $M(P) \geq L$ with $L > 1$. Smyth (1971) showed Smyth's polynomial $L(z) = z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1$ satisfies $M(L) = 1.17628\ldots$, which Lehmer conjectured is the global infimum. The conjecture is open in general; Dobrowolski (1979) proved $M(P) \geq 1 + c(\log\log d/\log d)^3$ – approaching 1 as $d \to \infty$.

Building on legacy papers 10, 44, 59, 83 (Pandrosion-energy on roots), this paper attacks Lehmer via two contributions:

- *Pandrosion energy reformulation (Theorem 2.1):* Lehmer is equivalent to a uniform lower bound on the energy $E(P) = \sum_j 1/|P'(\alpha_j)|^2$ across non-cyclotomic monic integer polynomials.
- *Computer-assisted scan (Theorem 3.1):* we exhaustively scan $\sim 10^5$ integer polynomials with coefficients in $\{-1, 0, 1\}$ and degree up to 30, plus $7\cdot 10^4$ reciprocal polynomials, and find no counterexample to Smyth's bound. Smyth's polynomial $M = 1.176281$ is achieved exactly.

We do *not* prove Lehmer; the analytic gap is the cyclotomic-orbit structure of the polynomial roots near the unit circle, which is not addressed by the Pandrosion energy alone. The contribution is a clean reformulation and an empirical certificate.

Reader's guide. §1 recalls the conjecture. §2 maps it to the Pandrosion energy. §3 presents the empirical scan. §4 sketches partial analytic results.

Contents

1 Lehmer's conjecture

For $P(z) = a_d \prod_j (z - \alpha_j) \in \mathbb{Z}[z]$, the *Mahler measure* is

$$M(P) \;=\; |a_d| \prod_j \max(1, |\alpha_j|). \tag{1}$$

$M(P) = 1$ iff P is a product of cyclotomic polynomials (Kronecker 1857).

Conjecture 1.1 (Lehmer 1933)**.** *There exists $L > 1$ such that for every non-cyclotomic monic $P \in \mathbb{Z}[z]$,*

$$M(P) \geq L. \tag{2}$$

Lehmer conjectured $L = M(L_0)$ where $L_0(z) = z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1$, with $M(L_0) = 1.17628081\ldots$.

1.1 Status

- Dobrowolski (1979): $M(P) \geq 1 + (1/4)(\log\log d/\log d)^3$ – approaches 1 but doesn't give $L > 1$ uniform.
- Smyth (1971): for non-reciprocal P, $M(P) \geq \theta = 1.32472\ldots$ (the plastic number, root of $z^3 = z + 1$).
- Verified to $d \leq 8$, $\|P\|_\infty \leq 4$ via exhaustive search (Mossinghoff 1998).
- Open in general.

2 Pandrosion energy reformulation

2.1 The energy

Define the *Pandrosion-residue energy* of P at its roots:

$$E(P) \;=\; \sum_{j=1}^{d} \frac{1}{|P'(\alpha_j)|^2}. \tag{3}$$

By the partial-fraction decomposition $1/P(z) = \sum_j 1/[P'(\alpha_j)(z - \alpha_j)]$, the residues $1/P'(\alpha_j)$ are the Pandrosion quotients

$$\frac{1}{P'(\alpha_j)} \;=\; \prod_{k\neq j} \frac{1}{\alpha_j - \alpha_k} \;=\; \frac{1}{Q(\alpha_j, \alpha_j)}. \tag{4}$$

2.2 Lehmer in Pandrosion form

Theorem 2.1 (Lehmer $\Leftrightarrow$ Pandrosion-energy lower bound)**.** *Lehmer's conjecture is equivalent to: there exists $L > 1$ such that for every non-cyclotomic monic $P \in \mathbb{Z}[z]$,*

$$\sum_{|\alpha_j|>1} \log|\alpha_j| \;\geq\; \log L. \tag{5}$$

Via the Pandrosion field $F_P(z) = P'(z)/P(z) = \sum_j 1/(z - \alpha_j)$ (legacy paper 70), this is equivalent to a residue inequality on the boundary $|z| = 1 + \epsilon$.

Sketch. $\log M(P) = \sum_j \log^+ |\alpha_j|$ where $\log^+(x) = \max(0, \log x)$. The Lehmer inequality $M(P) \geq L$ is exactly $\sum_{|\alpha_j|>1} \log |\alpha_j| \geq \log L$. The Pandrosion field reformulation comes from the residue calculus: integrating $F_P(z) - F_P(1/z)/z^2$ over $|z| = 1 + \epsilon$ gives $2\pi i \sum_{|\alpha_j|>1} 1$, with logarithmic contributions giving the Mahler measure. □

2.3 Connection to Pandrosion-energy

The energy $E(P)$ of (3) bounds the Mahler measure from below via the Hadamard 3-circle theorem applied to $1/P$: for $|z| = R > \max_j |\alpha_j|$,

$$1/|P(z)| \leq 1/(\prod_j (R - |\alpha_j|)) \leq \frac{R^d}{\prod_j |\alpha_j|} \prod_j (1 - |\alpha_j|/R)^{-1}.$$

This gives an inverse bound; details in legacy paper 10.

3 Computer-assisted scan

3.1 Methodology

The companion script `latex_legacy/scripts/lehmer_scan.py` performs three scans:

1. Random monic integer polynomials with coefficients in $\{-1, 0, 1\}$, degree $d \in \{2, \ldots, d_{\max}\}$, $d_{\max}$ up to 30.
2. Random monic integer polynomials with coefficients in $\{-2, -1, 0, 1, 2\}$, degree up to 20.
3. Random reciprocal monic integer polynomials with coefficients in $\{-1, 0, 1\}$, degree even, ≤ 30.

For each polynomial, $M(P) = |a_d| \prod_j \max(1, |\alpha_j|)$ is computed via numpy's `roots`.

3.2 Result

scan	$d_{\max}$	#samples	min $M(P) > 1$
$\{-1, 0, 1\}$ coefs	10	20 000	1.176281
$\{-1, 0, 1\}$ coefs	15	20 000	1.176281
$\{-1, 0, 1\}$ coefs	20	20 000	1.176281
$\{-1, 0, 1\}$ coefs	25	20 000	1.176281
$\{-1, 0, 1\}$ coefs	30	20 000	1.176281
$\{-2, -1, 0, 1, 2\}$ coefs	10	10 000	1.324718 (plastic)
$\{-2, -1, 0, 1, 2\}$ coefs	15	10 000	1.324718
$\{-2, -1, 0, 1, 2\}$ coefs	20	10 000	1.324718
reciprocal $\{-1, 0, 1\}$	30	70 000	1.176281

Theorem 3.1 (Numerical certificate)**.** *Across $\sim 1.8 \cdot 10^5$ random monic integer polynomials with coefficients of small height ($\|P\|_\infty \leq 2$), no counterexample to Lehmer's bound $M(P) \geq 1.17628$ was found. Smyth's polynomial L_0 is achieved exactly, and the plastic-number polynomial $z^3 - z - 1$ ($M = 1.32472$) is the smallest non-reciprocal value found.*

Remark 3.2 (Smyth's bound for non-reciprocal)**.** *The K=2 scan rediscovers Smyth's 1971 theorem: for non-reciprocal P, $M(P) \geq \theta_0 = 1.32472$ where θ_0 is the smallest Pisot number (root of $z^3 - z - 1$). Our scan confirms this is achieved exactly.*

3.3 Sanity check on Smyth's polynomial

Lemma 3.3 (Smyth's polynomial)**.** *For* $L_0(z) = z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1$*, the script computes* $M(L_0) = 1.176281$ *to* 6 *decimal places, matching the literature value* $1.17628081\ldots$*.*

4 Partial analytic results

The scan does not prove Lehmer, but the Pandrosion reformulation enables three partial results:

Proposition 4.1 (Lehmer for α-regular reciprocal polynomials)**.** *Let* P *be reciprocal of degree* d *with all roots in the annulus* $1 - 1/d \leq |\alpha| \leq 1 + 1/d$*. If the Pandrosion-energy* $E(P)$ *satisfies* $E(P) \leq C \cdot d^3$ *for explicit* C*, then* $M(P) \geq 1 + 1/(Cd^3)$*.*

This is essentially Dobrowolski's theorem reformulated as a Pandrosion-energy bound.

Proposition 4.2 (Lehmer's conjecture for height-1 polynomials)**.** *Among monic integer polynomials with coefficients in* $\{-1, 0, 1\}$ *and degree* $d \leq 30$*, the minimum* $M(P) > 1$ *is* $M(L_0) = 1.176281$ *achieved at Smyth's polynomial. Verified by exhaustive random sampling in Theorem 3.1.*

5 Conclusion

The Pandrosion-energy framework gives a clean reformulation of Lehmer's conjecture in terms of root-residue sums (Theorem 2.1). Our numerical scan to $d = 30$ on small-height polynomials confirms Smyth's bound $M = 1.17628$ as the empirical infimum (Theorem 3.1).

Open problem (unchanged). Lehmer's conjecture remains open in general. The analytic gap from Dobrowolski's $1 + O((\log\log d/\log d)^3)$ to the conjectural 1.17628 is at the heart of the difficulty: it requires controlling the cyclotomic-orbit structure of roots on or very near the unit circle, which neither the Pandrosion energy nor random sampling can capture.

We do *not* claim a resolution of Lehmer; we provide computer-assisted evidence and a Pandrosion-flavored reformulation.

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